

W-1. Irrigation Canal Improvement				
Water Resources Development	Medium	High		
Measure Type	Cost	Preferences by Community	Hazards	Implementation Time of Measures

Descriptions									
Measures	Improving irrigation canals in hilly regions is a vital strategy for enhancing agricultural productivity and promoting sustainable land management. By emphasizing careful site analysis, careful design, community engagement, and sustainable practices, we can develop efficient irrigation systems that are beneficial to farmers and the environment. By working together, we can create long-lasting irrigation systems that support the sustained growth of hilly region.								
	In the context of the CHT, irrigation canals are an essential conservation strategy that can protect the ecosystem and increase agricultural output. It is possible to create resilient and efficient irrigation systems by emphasizing the implementation of best practices, community involvement, sustainable design considerations, and thorough site assessment. These initiatives will help protect the region's abundant natural resources and support local livelihoods while maintaining a healthy balance between ecology and agriculture in the face of growing environmental problems.								
Location	R-114	R-99	R-65	K-98	K-76	K-71	B-173	B-77	B-76
	■	■	■	■	■	■	■	■	■
Applicable Sites	• Terraced or Contour-Aligned Land • Near Perennial Water Sources (Rivers, Springs, Reservoirs) • Valley Bottoms and Low-Lying Areas								
Slope	Less than 3%		3% to 15%		15% to 30%		30% to 60%		Above 60%
	√		√						
Design options and performance	• Gravity-Fed Contour Canals, Suitable for Slopes between 2-8%. • Lined Canals (Concrete, HDPE, or Stone Masonry), suitable for Steep slopes (above 5%) to reduce seepage and erosion. • Step-Down Canal with Drop Structures suitable for Slopes above 8%. • Terraced Irrigation Systems suitable for Sloping agricultural land.								
Feasibility criteria	Technical Feasibility • Ensures slope stability, proper soil conditions, and adequate water availability for efficient irrigation without excessive erosion or seepage. Environmental Feasibility • Prevents erosion and land degradation through sustainable water flow management. • Maintains ecological balance by minimizing habitat disruption. • Reduces wastage and promotes efficient water use in hilly terrains.								

	Economic Feasibility - Balances affordability and durability (e.g., earthen vs. lined canals). - Increases agricultural productivity, improving farmers' incomes. - Uses locally available materials and community-based management to reduce upkeep costs. Social Feasibility - Engages local farmers in planning, implementation, and maintenance. - Enhances food security and economic opportunities. - Aligns with government policies and local governance structures for long-term sustainability.			
Operation and maintenance	Routine Maintenance – Desilting, vegetation control, erosion repair. Structural Repairs – Crack sealing, check dam rehabilitation, slope stabilization. Seasonal Adjustments – Monsoon embankment reinforcement, dry-season storage optimization. Community Involvement – Water user groups, local government support, training programs.			
Cost	Medium			
Benefits	Benefits are as follows-			
	Type	Low	Medium	High
	Canopy Cover Increase			
	Carbon Sequestration			
	Crop Production Increase			
	Crop Failure Risk Reduction			
	Ensuring Food Security			
	Fruit Production Increase			
	Forestry Goods Production Increase			
	Livestock Production Increase			
	Fisheries Production Increase			
	Alternative Livelihood Generation			
	Increase of Income			
	Soil Erosion Control			
	Improved Soil Health			
	Soil Fertility Increase			
	Sustainable Water Supply			
	Water Quality Improvement			
	Water Purification			
	River Bank Erosion Control			
	Surface Runoff Reduction			
	Groundwater Recharge			
	Pest and Disease Management			
	Reduction of Pesticides and Fertilizer Use			
	Ecosystem and Biodiversity Increase			

	Better Communications			
	Better Education			
	Better WASH facilities			
Implementation Challenges				
	Type	Low	Medium	High
	Land unavailable/high land area required			
	Supply chain disconnection			
	Land right related problems			
	High human investment capital need			
	Insufficient infrastructures			
	Less demand of agroforestry product			
	Environmentally unsustainable			
	Socially unacceptable			
	Less beneficial			
Environmental performance	• Soil Erosion Control – Reduces sediment loss and stabilizes slopes. • Water Conservation – Prevents excessive runoff and promotes groundwater recharge. • Carbon Footprint Reduction – Encourages sustainable water management practices over energy-intensive irrigation methods. • Ecosystem Balance – Protects biodiversity by maintaining a natural water flow pattern and preventing habitat destruction. • Climate Resilience – Enhances water availability during droughts and controls floods during heavy rainfall.			
Remarks	Each canal system should be designed to match the unique characteristics of the area, including slope, soil type, and water availability. The long-term success of these systems relies heavily on active community participation, particularly through the engagement of local farmers in operation and maintenance. Additionally, integrated water management is essential, ensuring that proper drainage planning accompanies irrigation development to prevent issues like waterlogging and land degradation. Sustainability must be a priority in future projects, incorporating nature-based solutions such as vegetative buffer zones to enhance ecological benefits and long-term resilience.			